Functions

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
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Functions

“*Temporary* mini-world”

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message = greeting("Mogens")
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Functions

"*Temporary* mini-world"

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message = greeting("Mogens")
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```

Miniopgave

Skriv en funktion der returnerer kvadratet af et tal:

`square(4)`

(skal returnere 16)

Miniopgave

Skriv en funktion der returnerer summen af kvadrater
(sum of squares)

```
sum_of_squares(4, -2, 3)
```

(skal returnere 29)

```python
def square(v):
    return v**2

def sum_of_squares(a, b, c):
    a_sq = square(a)
    b_sq = square(b)
    c_sq = square(c)
    return a_sq + b_sq + c_sq

ssq = sum_of_squares(4, -2, 3)
print(ssq)
```

```python
def square(v):
    return v**2

def sum_of_squares(a, b, c):
    return square(a) + square(b) + square(c)

ssq = sum_of_squares(4, -2, 3)
print(ssq)
```

Funktionenskaldets midlertidige og private verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

“*Temporary* mini-world”

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

“*Temporary* **mini-world**”

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

"*Temporary* mini-world"

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

“*Temporary* mini-world”

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionenskaldets midlertidige og private verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionens private midlertidige verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

`salute` is private to the function

Funktionens private midlertidige verden

```python
def greeting(name):
    salute = "Hello"
    return salute + " " + name

salute = "Bonjour"
message = greeting("Mogens")
print(message)
print(salute)
```

`salute` is *still* private to the function

Funktionens private midlertidige verden

Python has to look for **`salute`** *outside* the function. Does it find it?

```
def greeting(name):
    # salute = "Hello"
    return salute + " " + name

salute = "Bonjour"
message = greeting("Mogens")

print(message)
```

Funktionens private midlertidige verden

Python has to look for **`salute`** *outside* the function. Does it find it?

```
def greeting(name):
    # salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
salute = "Bonjour"
print(message)
```

Variable og funktioner

- Et funktionskald skaber en midlertidig verden som forsvinder igen når funktionen returnerer en værdi.
- Alle variable man giver værdier i en funktion er private for funktionen.
- Hvis en variabel ikke er defineret i en funktion vil Python prøve at finde en definition udenfor funktionen.

Hvad er galt?

```
def greeting(name):
    name = ""
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Hvad er galt?

```
def greeting(nama):
    salute = "Hello"
    return salute + " " + name

name = "Henning"
message = greeting("Mogens")
print(message)
```

Builtin functions

```
abs_val = abs(-42)
print(abs_val)

fruit = "Banana"
length = len(fruit)
print(length)

print(type(length))
print(type(fruit))
```

https://docs.python.org/3.8/library/functions.html

Python 3.9 dokumentation

https://docs.python.org/3.9/

Look for "Library Reference"

Miniopgave

Skriv en funktion der tester om et tal er lige eller ulige

```
is_even(42)
```

skal returnere `True`

```
is_even(17)
```

skal returnere `False`